

MediTrack AI — Final Project Portfolio

MediTrack AI

AI-Powered Healthcare Management System

A comprehensive Android application connecting Patients, Doctors, and Admins with AI-assisted health analysis, appointment management, and smart reminders.

Android

Java

Firebase

Groq AI

v1.0

About the App

MediTrack AI is a full-featured Android healthcare management application built for three distinct user roles: **Patients**, **Doctors**, and **Admins**. The app integrates Firebase Realtime Database and Firebase Authentication for secure, real-time data handling, while leveraging the **Groq AI API** to provide intelligent symptom analysis and health recommendations.

Patients can book appointments, manage their medicine schedule with reminders, view prescriptions, and consult an AI chatbot for preliminary health guidance. Doctors can manage their appointments, write digital prescriptions with auto-generated PDFs, and review patient histories. The Admin panel provides full oversight of all users, appointments, prescriptions, and chat logs.

App Name	MediTrack AI
Version	1.0
Platform	Android (minSdk 24 / targetSdk 34)
Language	Java
IDE	Android Studio
Database	Firebase Realtime Database
Auth	Firebase Authentication + Google Sign-In
AI API	Groq API (via Retrofit)
Build System	Gradle (Kotlin DSL)

App Features

Feature Category	Description
■ Authentication	Email/password login, Google Sign-In, phone OTP, role-based registration (Patient / Doctor / Admin)
■ AI Health Analyzer	Groq AI-powered symptom analysis and doctor recommendation engine with chat interface
■ Appointments	Patients book appointments; doctors accept/decline; full history for both roles
■ Prescriptions	Doctors write digital prescriptions with medicines; auto-generated PDF sent to patients
■ Medicine Reminders	Scheduled exact alarms for medicine intake with local notifications
■ Medicine Inventory	Patients manage their medicine stock with quantity tracking
■ Push Notifications	Firebase Cloud Messaging for real-time appointment and prescription alerts
■ Profile Management	Edit profile, change password, upload profile photo via Firebase Storage
■ ■ Admin Panel	Manage all users, doctors, patients, appointments, prescriptions, and chat logs
■ Adherence Tracking	Log and track medicine adherence history per patient

■ ■ Technologies Used

Category	Technology / Library
Language	Java
IDE	Android Studio (Ladybug)
UI Layout	XML, ViewBinding, DataBinding
Architecture	MVVM (ViewModel + LiveData)
Navigation	Android Navigation Component (Safe Args)
Auth	Firebase Authentication, Google Sign-In API
Database	Firebase Realtime Database

Storage	Firebase Storage
Notifications	Firebase Cloud Messaging (FCM)
AI Integration	Groq API via Retrofit2 + OkHttp3
Image Loading	Glide
Animations	Lottie
PDF Generation	iText7 Core
Build	Gradle 8.7 (Kotlin DSL), compileSdk 34

■ APK Installation Guide

APK Installation on Android Device:

1. **Step 1:** Download the APK file from the GitHub repository.
2. **Step 2:** On your Android device, go to Settings > Security.
3. **Step 3:** Enable "Install from Unknown Sources" (or "Allow this source" when prompted).
4. **Step 4:** Open the downloaded APK file using a file manager.
5. **Step 5:** Tap Install and wait for completion.
6. **Step 6:** Open MediTrack AI and create your account.
- 7.

■ Running the Project Guide

Requirements

- Android Studio
- Firebase Project
- Groq API Key
- Android SDK (API Level 24+)

Steps

1. Clone the repository.git clone [https://github.com/\[YOUR_USERNAME\]/MediTrackAI.git](https://github.com/[YOUR_USERNAME]/MediTrackAI.git)
2. Open the project in Android Studio.
3. Replace the existing google-services.json file with your Firebase configuration file.
4. Sync Gradle files.
5. Add your Groq API Key inside:RetrofitClient.java
6. Build and run the application on an Android emulator or physical device.

■ Privacy Policy

Effective Date: June 2026

App Name: MediTrack AI

Developers: 1.NOMAN HAMMEED 2. MUHAMMAD SAAD

1. Information We Collect

MediTrack AI collects the following personal information when you register and use the application:

- Full name, email address, phone number, and password.
- Medical information including symptoms, prescriptions, and appointment history.
- Profile photos uploaded by the user.
- Device token for push notification delivery via Firebase Cloud Messaging.

2. How We Use Your Information

Your data is used solely to provide and improve the MediTrack AI service:

- To authenticate your identity and manage your account.
- To connect patients with doctors and manage appointments.
- To deliver medicine reminders and health notifications.
- To generate AI-powered health analysis through the Groq API.
- To generate prescription PDF documents for patients and doctors.

3. Data Storage and Security

All user data is stored securely on Google Firebase servers with industry-standard encryption. We implement role-based access control so that each user can only access data relevant to their role. Passwords are managed through Firebase Authentication and are never stored in plain text.

4. Third-Party Services

This app integrates the following third-party services, each with their own privacy policies:

- **Google Firebase:** Authentication, Realtime Database, Storage, Cloud Messaging.
- **Groq API:** AI language model for symptom analysis (data is sent anonymously).
- **Google Sign-In:** Optional login method using your Google account.

5. Data Sharing

We do not sell, rent, or share your personal information with any third parties for commercial purposes. Data is only shared between patients, doctors, and admins within the app as required for its core functionality.

6. Your Rights

You may request deletion of your account and all associated data at any time by contacting the developer. You may also update your personal information from within the app's Profile section.

7. Children's Privacy

MediTrack AI is not intended for use by children under the age of 13. We do not knowingly collect personal information from minors.

8. Contact

For privacy-related questions, please contact: (gxnoman235@gmail.com)(msaad17060@gmail.com).

■ Developer Information

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